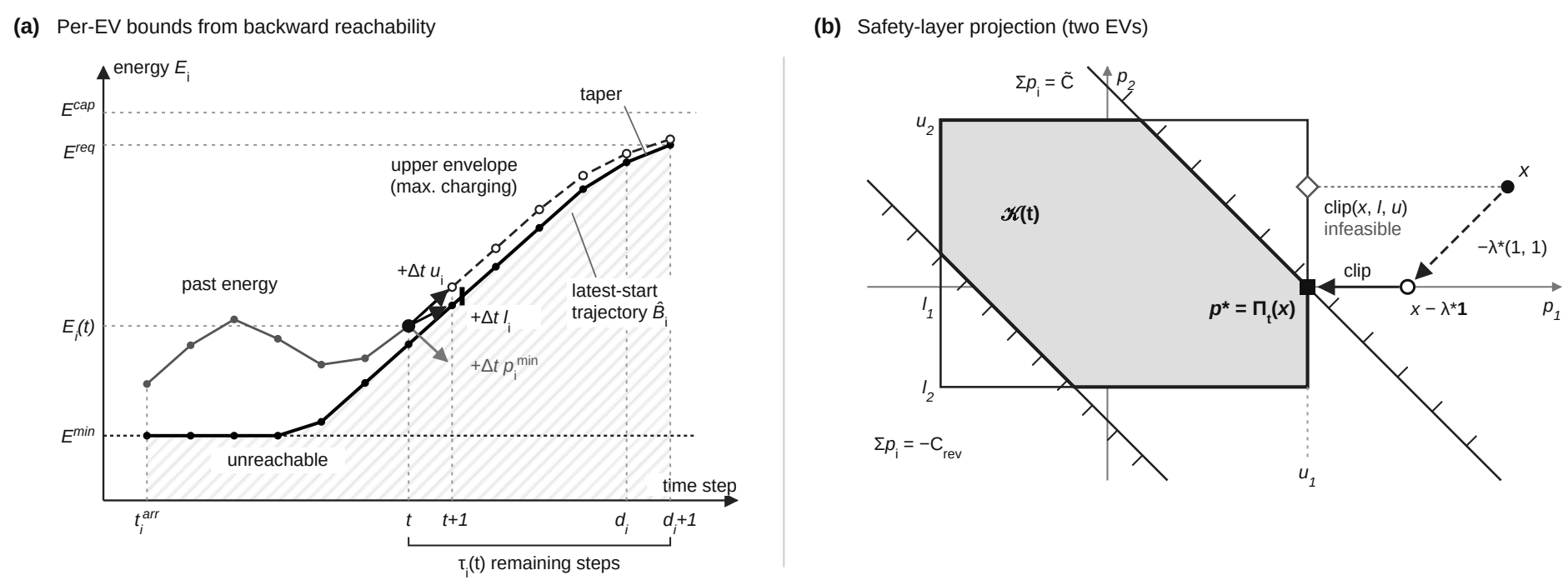

(a) Per-EV bounds from backward reachability
energy E_i
E^{cap}
E^{req}
taper
upper envelope
(max. charging)
past energy
$+\Delta t\ u_i$
$+\Delta t\ l_i$
$+\Delta t\ p_i^{min}$
latest-start
trajectory $\hat{B}_i$
$E_i(t)$
E^{min}
unreachable
time step
t_i^{arr}
t
$t+1$
d_i
d_i+1
$\tau_i(t)$ remaining steps
(b) Safety-layer projection (two EVs)
$\Sigma p_i = \tilde{C}$
p_2
u_2
$\mathcal{K}(t)$
l_1
l_2
$\Sigma p_i = -C_{rev}$
u_1
x
clip(x, l, u)
infeasible
$-\lambda^*(1, 1)$
clip
$p^* = \Pi_t(x)$
$x - \lambda^*\mathbf{1}$
p_1